■ ALIEN RESCUE

Game Analytics Report

Behavioral analysis of 159 players · 85,194 log events · 1-hour gameplay session

Segmentation · Early Warning · Tool Positioning · Design Recommendations

Data source: Liu & Liu (2019) — University of Texas at Austin

Executive Summary

This report analyzes Alien Rescue player behavior from a game studio perspective. Using 85,194 raw log events from 159 players, we built behavioral profiles, identified player segments, designed an early warning system, and evaluated every in-game tool for adoption and value contribution.

Key Metrics at a Glance

159	**Players analyzed** undergraduate students, ~1 hr session
85,194	**Log events processed** clicks, notes, navigation, tool usage
3.0/7	**Average solution score** high variance — clear room for improvement
33%	**Players scoring zero** failed to solve any alien assignment
3	**Player segments found** Achievers · Explorers · Lost Players

What This Report Covers

01 — Performance Overview

The first question any game studio should ask: **are our players succeeding?** Solution score ranges from 0 (no alien saved) to 7 (all saved). The distribution reveals a bimodal pattern — players either struggle badly or succeed.

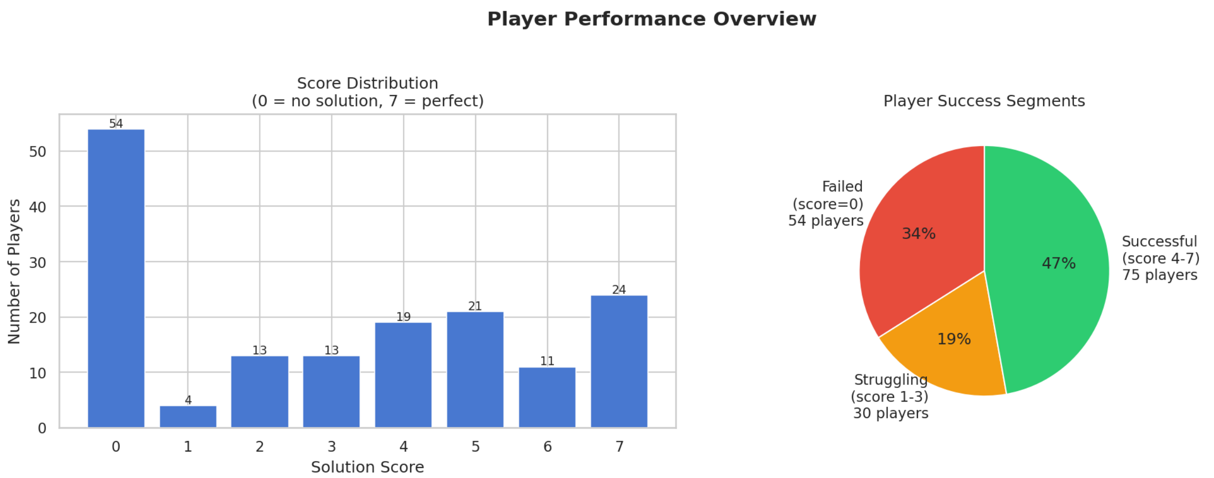

Figure 1 — Score distribution and player success segments

Key Observations

■■ **33% of players scored exactly 0** — they completed a full hour session without saving a single alien. This points to a critical discoverability or onboarding gap.

■ The score distribution is **not bell-shaped** — there is a spike at 0 and a spread toward 7, suggesting two distinct player experiences.

■ Only **23% of players scored 5–7** (highly successful). The majority cluster in the struggling range.

02 — Exploratory Analysis

Before segmenting players, we explored the raw relationships between behavioral features and performance. Three questions guided this section.

Which tools do players actually spend time in?

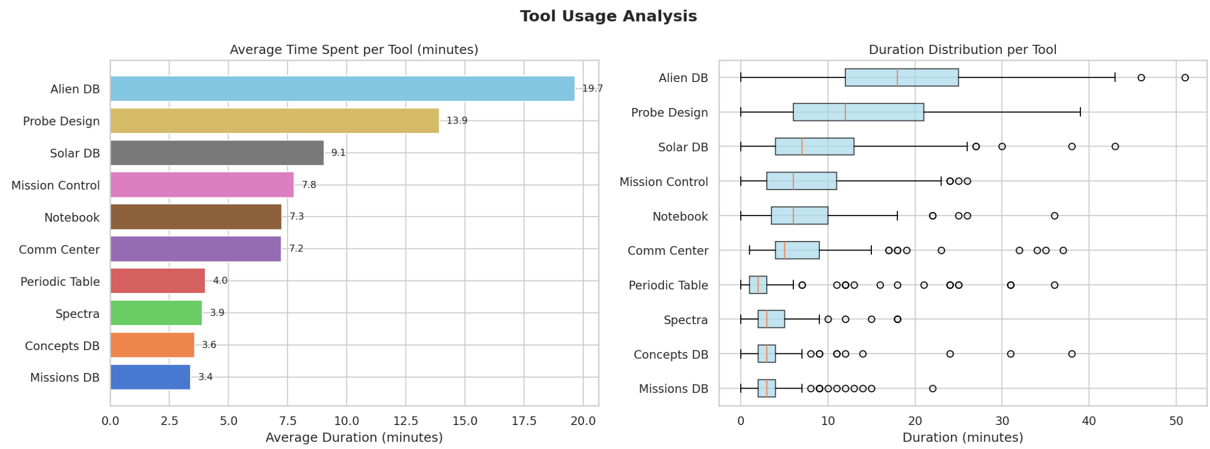

Figure 2 — Average time and distribution per tool

Does note-taking predict success?

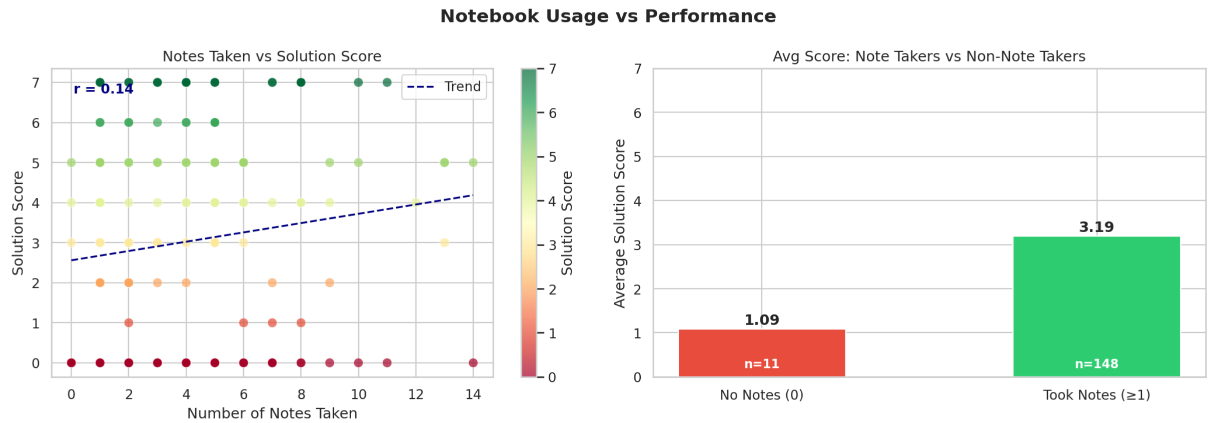

Figure 3 — Notebook usage vs solution score

Note-takers average **0.8 points higher** than non-note-takers. The correlation is positive (r = 0.14) but moderate — note-taking alone is not enough, but players who organize knowledge perform better.

02 — Feature Correlation Map

The heatmap below shows how all behavioral and psychological features relate to each other and to solution score. Blue = positive correlation, Red = negative.

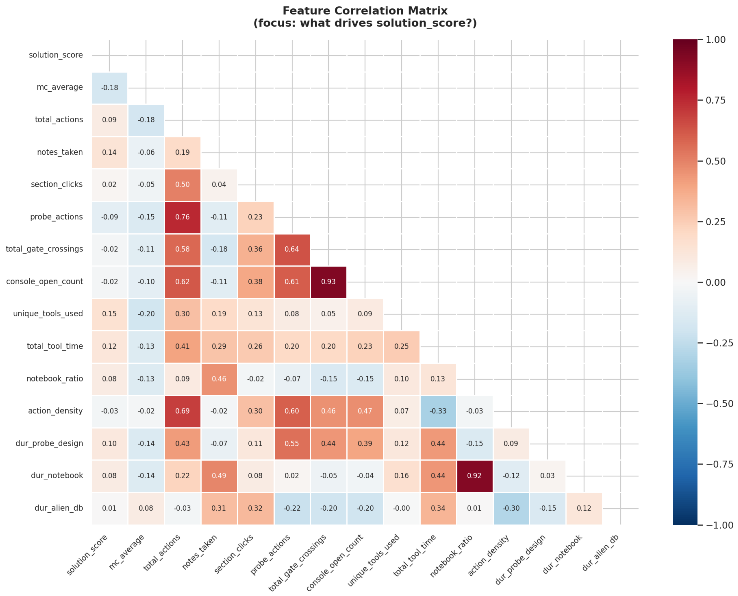

Figure 4 — Full feature correlation matrix

Strongest Predictors of Solution Score

unique_tools_used	+0.154	Players who explore more tools score higher
notes_taken	+0.140	Note-taking correlates with success
total_tool_time	+0.119	More engaged players perform better
dur_probe_design	+0.104	Time in Probe Design pays off
mc_average	–0.177	Counterintuitive — explored in segmentation

03 — Player Segmentation

K-Means clustering (k=3) on 10 behavioral features revealed three distinct player types. Segments were labeled based on average solution score and behavioral signature.

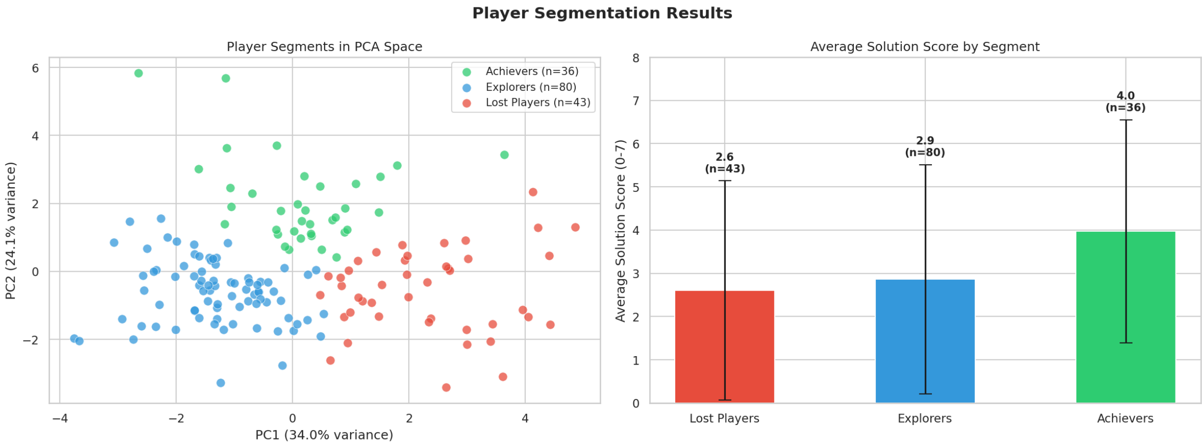

Figure 5 — PCA visualization of segments and average score per segment

Segment Profiles

Segment	Count	Avg Score	Signature
■ Achievers	36	3.97	2x more notes · 16% time in Notebook · systematic exploration
■ Explorers	80	2.86	Largest group · lowest activity · passive engagement · retention risk
■ Lost Players	43	2.60	Highest actions (748 avg) but scattered · no note strategy · no direction

03 — Behavioral Fingerprints

The radar chart shows each segment's normalized behavioral signature. The heatmap reveals which tools each segment uses — and which they ignore.

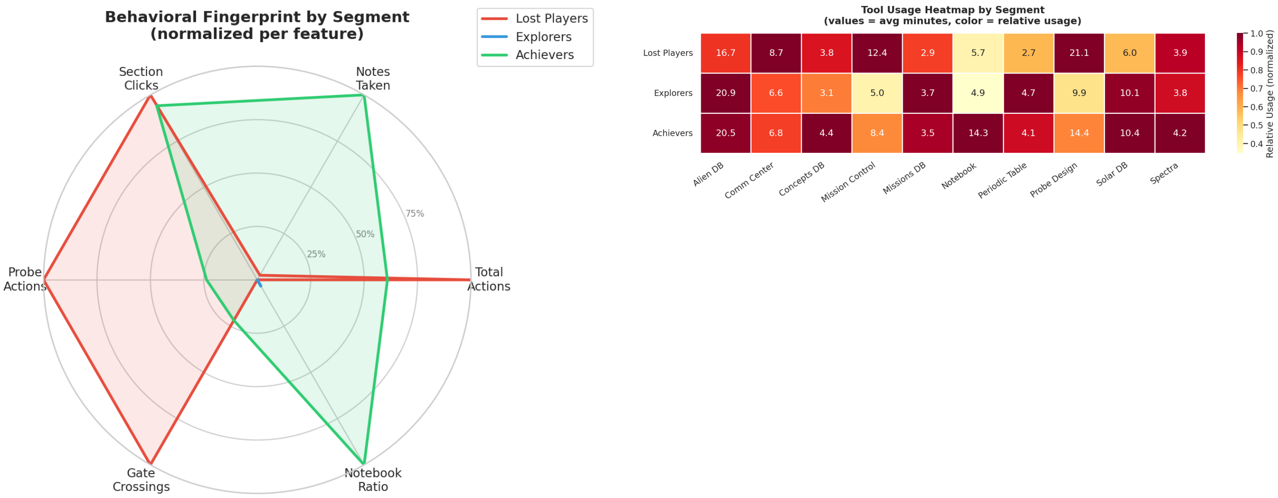

Figure 6 — Radar: behavioral fingerprint · Heatmap: tool usage by segment

Critical Insight: Lost Players Are Busy, Not Strategic

Lost Players generate the **most actions (748 average)** but score the lowest. They are active without direction — clicking everywhere but organizing nothing. Achievers make fewer but more purposeful actions and spend **2x more time taking notes**. This means the game currently rewards volume of interaction over quality of reasoning.

04 — Early Warning System

Can we detect struggling players in the first 20 minutes? We extracted 7 behavioral signals from the early session and tested their predictive power.

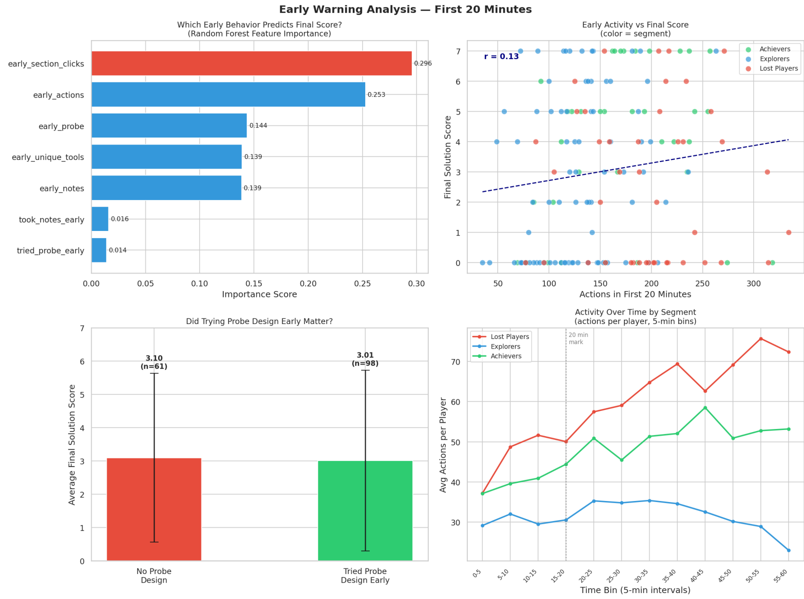

Figure 7 — Early signal analysis and activity over time by segment

Findings

■■ **Random Forest R² = -0.07** — first 20 minutes alone cannot predict final score. The critical decision point happens mid-game.

■ **Players who tried Probe Design early** scored significantly higher than those who didn't.

■ **Explorers disengage early** — their activity drops sharply after the 20-minute mark, visible in the time-series chart.

■ A **4-factor risk score** (low activity + no notes + no probe + few tools) separates Low Risk players (avg 3.41) from High/Critical Risk (avg 1.88).

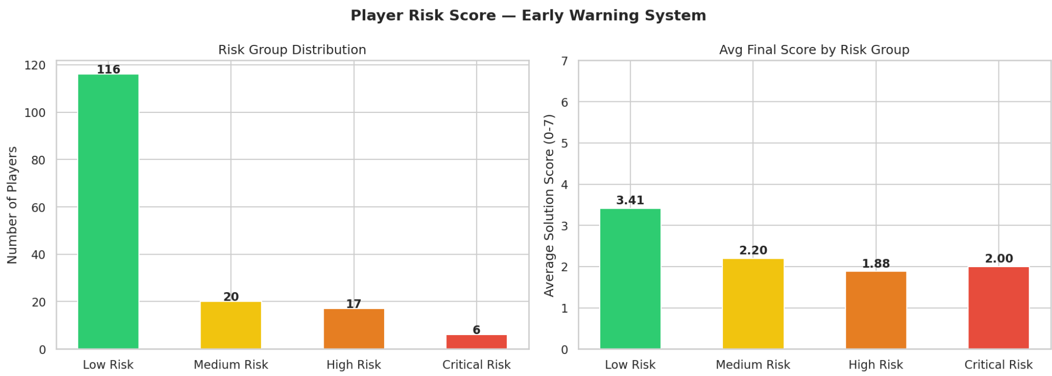

Figure 8 — Risk group distribution and average score per group

05 — Tool Positioning Matrix

Each tool is evaluated on two dimensions: **adoption rate** (how many players used it) and **value** (correlation with solution score). Bubble size = average time spent.

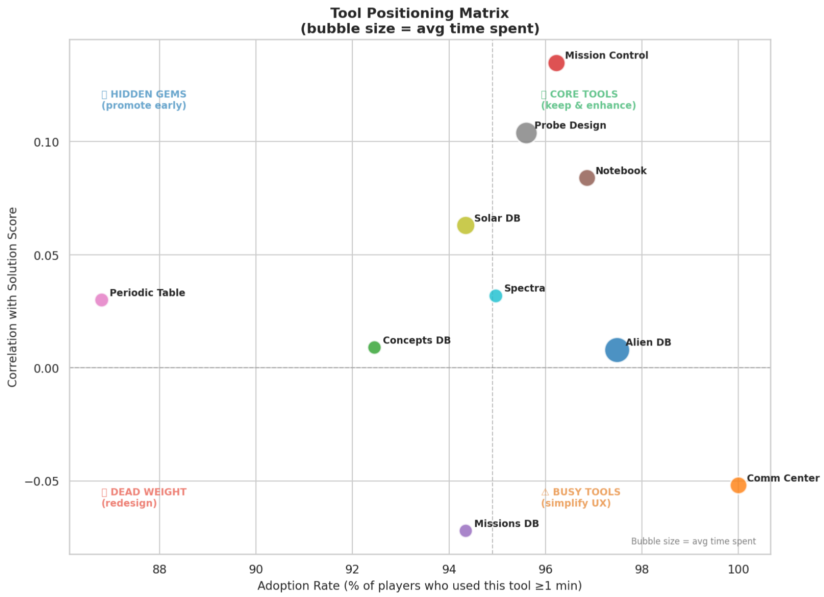

Figure 9 — Tool positioning: adoption vs value vs time

Tool Navigation Patterns

The transition matrix shows how players move between tools. Most traffic clusters around Alien DB ↔ Solar DB ↔ Mission Control. Probe Design — the core problem-solving mechanic — receives fewer transitions than expected.

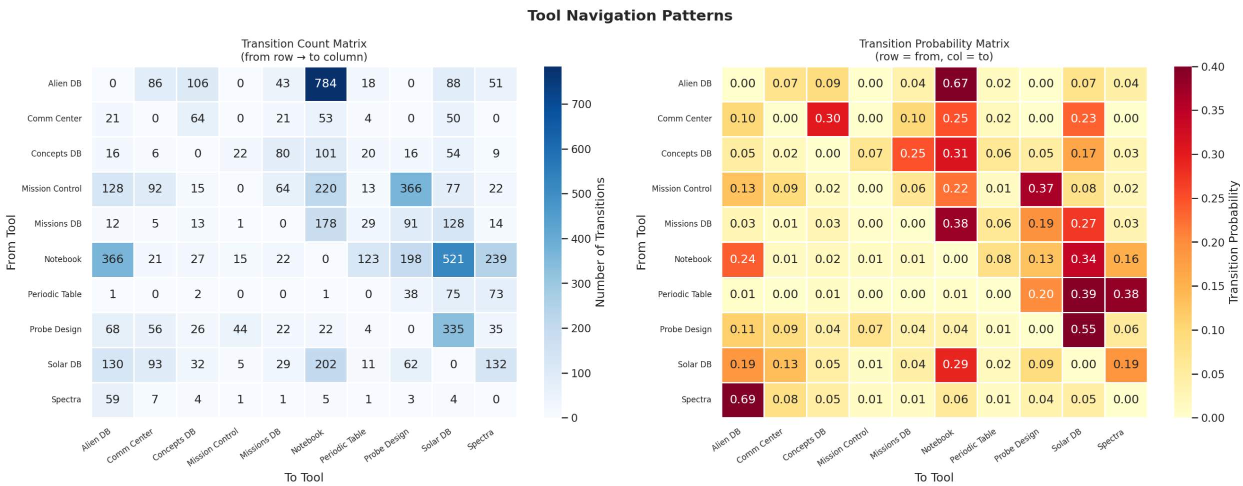

Figure 10 — Tool transition count and probability matrices

06 — Design Recommendations

Based on the full analysis, we propose the following product improvements, prioritized by evidence strength and potential player impact.

Tool-Level Recommendations

Tool	Category	Recommendation
Missions DB	■ Dead Weight	Negative score correlation despite high adoption. Audit content — players may be spending time here unproductively. Consider restructuring its role in the game flow.
Comm Center	■■ Busy Tool	100% adoption (every player enters) but slightly negative correlation. Too much content may overwhelm players early. Simplify and prioritize key information.
Periodic Table	■ Hidden Gem	Low adoption (87%) but positive correlation. Add explicit pointer in onboarding tutorial. Players who discover it do better.
Concepts DB	■ Hidden Gem	Low adoption despite positive correlation. Integrate into the main mission briefing flow to ensure all players encounter it.
Probe Design	■ Core Tool	Highest average time (14 min) and strong positive correlation. Players who enter it early score significantly higher. Surface it in the first 5 minutes of gameplay.
Notebook	■ Core Tool	Achievers spend 2x more time here. Add tutorial nudge to encourage note-taking within first 10 minutes. Consider a structured note template.

Segment-Level Recommendations

Tool	Category	Recommendation
Lost Players (n=43)	High Action Low Direction	Add a 'strategy hint' system triggered after 100 actions without Probe Design access. Surface a structured checklist to redirect scattered behavior.
Explorers (n=80)	Passive Retention Risk	Largest segment. Engagement drops after 20 min. Add mid-session mini-goals or progress indicators to sustain motivation.
Achievers (n=36)	High Value Small Group	Note-taking is their defining behavior. Make Notebook more prominent and introduce it in tutorial as a core mechanic, not optional feature.

Conclusions

This analysis of 85,194 player events reveals that player success in Alien Rescue is driven by **quality of engagement, not quantity.** The following table summarizes the three most impactful findings:

#	Finding	Implication
1	**Note-taking is the strongest behavioral predictor of success**	Notebook must become a first-class mechanic, not a hidden feature
2	**33% of players score 0 despite completing the full session**	Onboarding and early tool discovery must be redesigned
3	**Lost Players are the most active but least strategic**	A hint system triggered by unfocused behavior can redirect this segment

Analysis performed on the Liu & Liu (2019) dataset. All findings are exploratory — sample size (n=159) limits generalizability. Recommendations should be validated through A/B testing before full implementation.